Q-SCOPE: Coverage-Constrained Query-Adaptive Visual Token Pruning for Vision-Language Models

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Abstract

Large vision-language models answer visual questions from long visual-token sequences, and that prefix dominates prefill compute and the key-value cache. Existing pruners fall into three families, and each family leaves a distinct gap. Salience ranking keeps high-scoring tokens that still contain a large amount of similar content, and a score from one visual depth cannot represent what other depths emphasize. Diversity and coverage build a compact visual set, but image-level representativeness and the evidence required by the current question are not fully consistent. One-shot compression before the language model then freezes the visual length, so early interaction and later redundancy share one capacity. We formulate pruning as query-adaptive subset selection under a layered compute budget and a coverage-fidelity constraint: near a fixed-capacity visual basis, raise question relevance while limiting coverage distortion and the number of identity edits. The training-free framework Q-SCOPE builds that basis from two visual depths and regional quotas, adapts it with query-guided exchange, and contracts the nested set with question-to-visual attention. On LLaVA-1.5-7B, the completed early schedule removes 94.4% of the visual token-layer budget and retains 95.66% of the unpruned six-task relative mean. Cells without a matched measurement are left blank. The relative mean is not a per-question accuracy, and the budget ratio is not a measured speedup.

1. Introduction

Large vision-language models connect a visual encoder to a language decoder and extend instruction following to images and video. A standard image already yields hundreds of visual tokens, and high-resolution crops or video frames make the prefix longer. Standard self-attention grows quadratically with that length, and the cache grows with it. Studies of visual redundancy show that a compact subset can still support many understanding tasks, which makes training-free token pruning a direct route to lower inference cost.

Current pruners divide into three families, and we reveal three coupled limitations.

Salience and interaction scores keep high-scoring locations. FastV uses shallow language-model attention, SparseVLM scores visual tokens with image-related text and recycles information across layers, and FitPrune fits the attention pattern under a compute budget [1–3]. An individual score does not state whether two kept tokens stand in for each other, so high-score regions still contain a large amount of similar content. Visual-token salience is also depth-dependent. The attention a patch receives, and the representative role it plays, change from a middle layer to a deep layer. A selection that reads only one depth therefore misses complementary information emphasized at the other depth.

Diversity and coverage use feature differences or a representative relation. DART down-weights duplicates of a pivot, DivPrune casts selection as a max-min diversity problem, and SCoRe unifies salience and coverage as a weighted k-center [4–6]. These criteria improve image-level retention. They are not fully consistent with the evidence the question needs. Background or edge locations that are weakly related to the question still enter the kept set through feature difference or coverage contribution, and they occupy limited interaction capacity. The local text, attributes, and relations required by the answer are not kept first. The representative set therefore has to be adapted to the question, while the loss of the existing coverage relation stays under an explicit bound.

One-shot compression before the language model fixes the later visual length. VisionZip mixes dominant tokens with context, CDPruner puts instruction relevance into conditional diversity, and CoIn combines salience, cross-modal alignment, and coverage [7–9]. After that single cut, every later layer sees the same visual capacity. A small entry discards candidates before cross-modal interaction has formed. A large entry carries the extra tokens into deep layers, where they add little further accuracy and block additional compres-

sion. PyramidDrop, SparseVLM, and ViTCoP already shorten the sequence by stages [10, 11]. For a nested schedule, a location that never enters the first set cannot be restored by a later ranking. Entry identity and later capacity have to be chosen together.

We cast visual-token pruning as query-adaptive subset selection under a layered compute budget and a coverage-fidelity constraint. Near a fixed-capacity visual basis, the objective raises question relevance. The constraints limit total coverage distortion and the number of identity edits. Nested sets then contract the visual capacity across interaction stages. This statement links who is admitted to the decoder with how computation is allocated afterward, and it keeps representativeness, task relevance, and inference cost in one problem.

Q-SCOPE is a training-free instance of that problem. Dual-depth salience and descriptors build a regional representative basis. Query-guided exchange (QGE) then replaces a bounded number of identities at constant capacity and rejects an edit whose coverage loss exceeds the allowed distortion. In the decoder, question-to-visual attention (Q-Attn), the attention from question tokens to the current visual positions, rescores the survivors at scheduled cuts. The forward pass still gathers the original visual rows at the selected indices.

On LLaVA-1.5-7B, the completed early schedule removes 94.4% of the visual token-layer budget and retains 95.66% of the unpruned six-task relative mean [12]. The same-budget comparison with other methods is reserved in Table 1 and is not filled from numbers measured under a different protocol.

Our contributions are threefold. First, we reveal three coupled limitations of visual token pruning: depth-dependent salience, the mismatch between visual coverage and question evidence, and a fixed capacity across interaction stages. Second, we formulate coverage-constrained query-adaptive pruning and instantiate it with dual-depth bases, regional quotas, QGE, and Q-Attn refinement. Third, we report the completed six-task result, the exchange trade-off, and the effect of when the first cut occurs, and we leave unmeasured cells blank.

2. Related Work

2.1. Large vision-language models

LLaVA and BLIP-2 connect a visual encoder to a language model through a projection or a query interface [12, 13]. Higher-resolution and video models, including LLaVA-NeXT, LLaVA-OneVision, and Qwen2.5-VL, lengthen the visual sequence with multi-scale crops, dynamic resolution, or multiple frames [14–16]. The longer prefix increases prefill cost and the key-value cache. This paper studies training-free pruning that chooses the entry set and then contracts it inside the decoder.

2.2. Visual token pruning

Salience methods show that an internal importance signal can drop many visual tokens. They do not, by themselves, describe duplication inside the kept set, and a single visual depth does not carry the ranking of another depth. We estimate both salience and the representative relation from position-aligned middle and deep features.

Coverage and diversity methods make the set-level decision explicit. A purely visual representative relation is not fully consistent with the question: a weakly related region still takes a slot when it is locally distinctive. Q-SCOPE starts from that visual basis and then edits it under a coverage floor.

Pre-decoder compression improves the entry, then holds its length fixed. A small entry removes evidence before interaction. A large entry spends the same capacity in deep layers. Layered schedules separate those two choices, but a delete-only schedule cannot recover a token that missed the entry. Q-SCOPE therefore builds the basis, adapts it with QGE, and refines the survivors with Q-Attn.

3. Method

3.1. Problem

Given an image or a video I and a question Q, the encoder produces N visual positions $\mathcal{V}$. The decoder has L layers. Layer ℓ reads a visual set $S^{(\ell)}$. The visual token-layer budget and its layer average are

$$W = \sum_{\ell=1}^{L} |S^{(\ell)}| \le W_{\max}, \qquad \bar{K} = W/L. \tag{1}$$

W counts visual positions seen by each layer. It is not FLOPs, latency, or memory. Let S_0 be a question-free basis of size K, q_i a question score, and $C(S)$ a coverage of all original positions. Entry adaptation solves

$$\max_{S \subseteq \mathcal{V}} \sum_{i \in S} q_i \quad \text{s.t.} \quad |S| = K,\ |S \triangle S_0|/2 \le B,\ C(S) \ge C(S_0) - \delta. \tag{2}$$

B bounds the number of identity edits and δ bounds coverage distortion. The implemented search does not claim a global optimum of Eq. (2). It builds S_0, then accepts a bounded number of one-for-one exchanges that stay inside a coverage envelope. The selected indices gather the original projected rows. Concrete ca-

pacities belong to the experimental schedule, not to this definition.

3.2. Cross-depth descriptors

Middle and deep features F_i^{m} and F_i^{d} share patch indices. $\hat{f}_i^a = F_i^a/\|F_i^a\|_2$, and s_i^a is the sum of heads of CLS-to-patch attention. A shared weight $\beta \in (0,1)$ builds the selection descriptor and the fused salience,

$$z_i = \big[\sqrt{1-\beta}\,\hat{f}_i^{\mathrm{d}};\ \sqrt{\beta}\,\hat{f}_i^{\mathrm{m}}\big], \quad s_i = (1-\beta)s_i^{\mathrm{d}} + \beta s_i^{\mathrm{m}}. \tag{3}$$

The inner product $z_i^\top z_j$ is a weighted sum of the two depth-wise cosines. s_i sets the anchor order and z_i measures local complementarity. z_i is not the tensor sent to the language model.

3.3. Regional quotas

Anchors are the top $K-b$ positions by s. The residual pool is everything outside the top K, not the complement of the anchors:

$$H = \mathrm{Top}_{K-b}(s;\mathcal{V}), \qquad R = \mathcal{V} \setminus \mathrm{Top}_K(s;\mathcal{V}). \tag{4}$$

The grid is split into disjoint regions V_g. A region that contains anchors uses their normalized descriptor sum as a prototype p_g. With $g(i)$ the region of position i,

$$c_i = z_i^\top p_{g(i)}, \qquad u_i = s_i\big[1 - \mathrm{clip}(c_i, 0, 1)\big]. \tag{5}$$

Only candidates with a valid prototype and $c_i > 0$ enter the utility sort. Each region contributes at most τ of them. If the quota cannot fill b seats, τ increases. Any remainder is filled by salience, preferring regions that already have anchors. The basis is $S_0 = H \cup D$. The quota is a count on the same utility, not a second coverage objective.

3.4. Coverage-constrained query-guided exchange

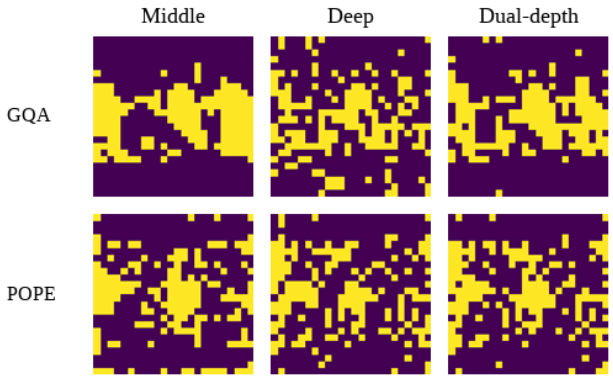


Figure 1. Cross-depth selections at one kept count. Yellow cells are retained patches. GQA and POPE are centered on their rows. Middle selections form larger blocks. Deep selections are more scattered.

Figure 1 is the same-capacity contrast behind the basis that QGE edits. Content words of the question, excluding options, instructions, and OCR lines, are averaged in the original embedding to $\bar{e}_Q$. For a native projected row x_i, the raw cosine is $a_i = \cos(x_i, \bar{e}_Q)$, and the question score is

$$q_i = \sigma\Big(\mathrm{clip}\big((a_i - \mathrm{med}(a))/(\kappa\,\mathrm{std}(a)), -c, c\big)\Big). \tag{6}$$

G_{ij} is the mean over views of the cosine clipped to $[0,1]$. Coverage and novelty are

$$C(S) = N^{-1}\sum_{i\in\mathcal{V}} \max_{j\in S} G_{ij}, \qquad \nu(v \mid S) = 1 - \max_{j\in S} G_{vj}. \tag{7}$$

Drop and donor priorities are $\pi_u^- = 1 - q_u + \lambda_r \max_{j\in S\setminus\{u\}} G_{uj}$ and $\pi_v^+ = q_v\nu(v \mid S)$. For $S' = (S\setminus\{u\})\cup\{v\}$, write $\Delta q = q_v - q_u$ and $\Delta C = C(S') - C(S)$. A feasible pair needs $\Delta q \ge \gamma$, $\nu(v \mid S) \ge \eta$, and $\Delta C \ge -\varepsilon$. Each round then solves

$$\max_{u,v}\ \Delta q\,\nu(v \mid S) + \lambda_c \Delta C. \tag{8}$$

The search stops when no pair passes the gates or B exchanges have been accepted. The result S_Q obeys

$$|S_Q \triangle S_0|/2 \le B, \qquad C(S_Q) \ge C(S_0) - B\varepsilon. \tag{9}$$

Choosing $\delta \ge B\varepsilon$ places this envelope inside Eq. (2). The envelope bounds a feature proxy, not answer correctness.

3.5. Question-to-visual attention refinement

At cut ℓ_j the schedule asks for a nonincreasing size K_j, starting from $R_0 = S_Q$. Q-Attn reads the question-token queries and the keys of the legal prefix under the model mask M,

$$A_{t,k}^{\ell_j,h} = \mathrm{softmax}_k\big((q_t^{\ell_j,h})^\top k_k^{\ell_j,h}/\sqrt{d_h} + M_{t,k}\big). \tag{10}$$

Let $p_j(i)$ be the current-sequence position of original index i. Summing over heads and question positions $\mathcal{T}_Q$ gives the keep score, and the next set is the top K_j survivors of the previous set. Retained tokens keep their original position ids. If capacity K_j lasts for n_j layers, then $W = \sum_j n_j K_j$. The entry decides which evidence can participate. Each cut decides how far that evidence continues.

4. Experiments

4.1. Setup

Image benchmarks are GQA, VizWiz, ScienceQA-IMG, TextVQA, POPE, MME, MMBench, MMMU,

and AI2D [17–25]. Video benchmarks are MLVU, LongVideoBench, and Video-MME [26–28]. POPE is F1. The six-task relative mean uses MME perception with denominator 1510.70, and GQA 61.90, POPE 85.80, MMBench 64.10, ScienceQA-IMG 69.51, and TextVQA 58.20. AI2D, MMMU, and VizWiz stay blank until their scores and denominators are recorded together. The main model is LLaVA-1.5-7B. Transfer tables reserve LLaVA-NeXT-7B, Qwen2.5-VL-7B, and LLaVA-OneVision-7B.

The seven comparison methods are SparseVLM and ViTCoP for salience and interaction, DivPrune and SCoRe for diversity and coverage, and AgilePruner, PruneSID, and CoIn for joint pre-decoder compression [29, 30]. Runs use NVIDIA V100 GPUs and lmms-eval at batch size one [31]. Layer-average capacity $\bar{K} = W/L$ aligns a flat entry with a layered schedule. On LLaVA-1.5-7B the three budgets are $\bar{K} \in \{128, 64, 32\}$, which remove 77.8%, 88.9%, and 94.4% of the 576×32 visual token-layer count. NeXT uses the same three ratios. Qwen2.5-VL-7B uses 80.2% and 90.1%. A dash means that cell has no measurement under this protocol. Published numbers from other papers are not copied into the table.

4.2. LLaVA-1.5-7B

At the 94.4% budget, regional quotas plus QGE score POPE 86.2, MME perception 1409.0, MMBench 61.6, TextVQA 53.3, ScienceQA-IMG 69.9, and GQA 57.1. ScienceQA-IMG and POPE stay near the unpruned references. GQA and TextVQA remain lower. The other two budgets and every external method are blank.

4.3. Transfer and video

Tables 2 and 3 use the six classic tasks. Each model is normalized by its own unpruned scores. Qwen is split into 80.2% and 90.1% removal. Both tables are unfilled. Video uses LLaVA-OneVision-7B on MLVU, LongVideoBench, and Video-MME at two retained lengths. Those cells are unfilled as well.

4.4. Efficiency

Efficiency has not been measured. The planned comparison, on one V100 with the same images and batch size 1, is the unpruned model, ViTCoP, SCoRe, CoIn, Q-SCOPE without QGE, and full Q-SCOPE. Accuracy runs stop naturally. Timing uses a fixed generation length so answer length does not enter the clock. Time to first token starts from the image and text already in memory and includes the vision tower, projection, basis, QGE, and prefill. The same run records generation time, peak memory, and the selector time. Warmup is excluded. Each setting is repeated, and we report the median, mean, and tail. Equal W only matches the visual token-layer count.

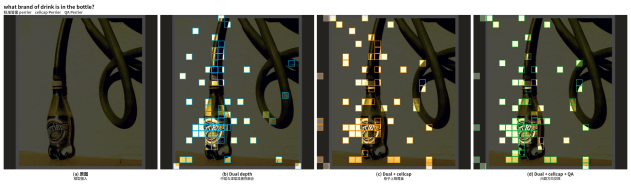


Figure 2. TextVQA bottle. Dim cells are dropped. From left to right: input, dual-depth, regional quota, and QGE. Both the quota basis and QGE answer Perrier.

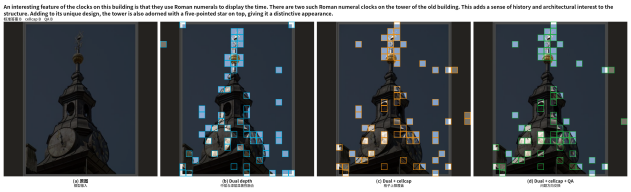


Figure 3. MMBench clock. The quota adds peripheral cells. QGE moves part of the selection back onto the clock face. Both answer B.

4.5. Ablations

With QGE off and MME as the perception-plus-cognition total, middle features lead GQA and POPE, deep features lead MMBench, and dual-depth plus the older coverage leads the four-task relative mean (Table 5). This protocol uses W=2048 and is not the 94.4% row above. On 1,024 images the mean Jaccard of the middle and deep top sets is 0.331. On MMBench, 35 questions are correct for the middle depth and for dual-depth but wrong for the deep depth, and 38 show the opposite pattern. Dual-depth supplies two rankings. It does not win every task.

Regional quotas are the strongest of these coverage variants, at 93.89% on the six-task mean in the unrounded record. Removing salience on empty cells drops POPE to 76.33. On that quota basis, the smallest QGE budget raises the six-task mean from 93.89% to 95.66%. GQA moves from 55.92 to 57.08, with 498 newly correct answers and 353 newly incorrect ones. POPE rises and TextVQA falls, from 53.85 to 53.28. Larger exchange budgets keep lifting POPE and lower the other tasks together. The same W=1024 with a late first cut scores 93.47%, against 87.67% and 87.45% for two earlier cuts. The late schedule has the smaller entry and the higher score. Changing only the last stage moves the six-task mean by 0.11 points, from 96.78% to 96.89%.

Table 1. LLaVA-1.5-7B. Rel. is the mean of per-task scores divided by the fixed references. The filled row is the regional-quota basis plus QGE under the early schedule, and its Rel. uses the six measured tasks. MME is perception.

Method	POPE	MME	MMB	MMMU	Text	AI2D	Viz	SQA	GQA	Rel.
Dense reference, 576 visual tokens										
LLaVA-1.5-7B	85.8	1510.7	64.1	–	58.2	–	–	69.5	61.9	100.0%
Layer average $K = 128$ (77.8% removed)										
SparseVLM [2]	–	–	–	–	–	–	–	–	–	–
ViTCoP [11]	–	–	–	–	–	–	–	–	–	–
DivPrune [5]	–	–	–	–	–	–	–	–	–	–
SCoRe [6]	–	–	–	–	–	–	–	–	–	–
AgilePruner [29]	–	–	–	–	–	–	–	–	–	–
PruneSID [30]	–	–	–	–	–	–	–	–	–	–
CoIn [9]	–	–	–	–	–	–	–	–	–	–
Q-SCOPE	–	–	–	–	–	–	–	–	–	–
Layer average $K = 64$ (88.9% removed)										
SparseVLM [2]	–	–	–	–	–	–	–	–	–	–
ViTCoP [11]	–	–	–	–	–	–	–	–	–	–
DivPrune [5]	–	–	–	–	–	–	–	–	–	–
SCoRe [6]	–	–	–	–	–	–	–	–	–	–
AgilePruner [29]	–	–	–	–	–	–	–	–	–	–
PruneSID [30]	–	–	–	–	–	–	–	–	–	–
CoIn [9]	–	–	–	–	–	–	–	–	–	–
Q-SCOPE	–	–	–	–	–	–	–	–	–	–
Layer average $K = 32$ (94.4% removed)										
SparseVLM [2]	–	–	–	–	–	–	–	–	–	–
ViTCoP [11]	–	–	–	–	–	–	–	–	–	–
DivPrune [5]	–	–	–	–	–	–	–	–	–	–
SCoRe [6]	–	–	–	–	–	–	–	–	–	–
AgilePruner [29]	–	–	–	–	–	–	–	–	–	–
PruneSID [30]	–	–	–	–	–	–	–	–	–	–
CoIn [9]	–	–	–	–	–	–	–	–	–	–
Q-SCOPE	86.2	1409.0	61.6	–	53.3	–	–	69.9	57.1	95.66%

Table 2. LLaVA-NeXT-7B. Six tasks. Scores are unfilled.

Method	MME	MMB	POPE	SQA	GQA	Text	Rel.
Dense	–	–	–	–	–	–	100%
77.8% / 88.9% / 94.4% removed							
ViTCoP	–	–	–	–	–	–	–
SCoRe	–	–	–	–	–	–	–
CoIn	–	–	–	–	–	–	–
Q-SCOPE	–	–	–	–	–	–	–

Table 3. Qwen2.5-VL-7B at 80.2% and 90.1% removal. Scores are unfilled.

Method	MME	MMB	POPE	SQA	GQA	Text	Rel.
Dense	–	–	–	–	–	–	100%
80.2% removed							
ViTCoP	–	–	–	–	–	–	–
SCoRe	–	–	–	–	–	–	–
CoIn	–	–	–	–	–	–	–
Q-SCOPE	–	–	–	–	–	–	–
90.1% removed							
ViTCoP	–	–	–	–	–	–	–
SCoRe	–	–	–	–	–	–	–
CoIn	–	–	–	–	–	–	–
Q-SCOPE	–	–	–	–	–	–	–

Table 4. Video. MLVU, LongVideoBench, and Video-MME are unfilled.

Method	MLVU	LVB	V-MME
Dense	–	–	–
ViTCoP	–	–	–
DivPrune	–	–	–
SCoRe	–	–	–
CoIn	–	–	–
Q-SCOPE	–	–	–

Table 5. Dual-depth, QGE off. MME-T is the total score. Rel.4 uses these four tasks only.

Guide	GQA	POPE	MME-T	MMB	Rel.4
Middle	57.9	84.0	1718.0	62.0	95.2
Deep	56.4	81.3	1745.7	63.0	94.6
Dual-depth	57.4	82.8	1744.5	62.6	95.3
Dual-depth + cover	57.2	83.0	1765.2	62.2	95.4

Table 6. Coverage variants, QGE off. MME is perception.

Rule	POPE	MME	Text	Rel.6
Previous cover	81.5	1339.3	53.5	93.5
Valid prototypes	81.5	1349.5	53.5	93.5
Drop empty-cell salience	76.4	1299.0	53.0	91.0
Update prototypes	82.2	1372.0	53.8	93.8
Regional quota	81.5	1376.4	53.9	93.9
Wider residual pool	81.1	1363.8	53.9	93.3

5. Conclusion

We formulate visual-token pruning as query-adaptive subset selection under a layered budget and a coverage-fidelity constraint, and we reveal three coupled limitations that this formulation separates: salience changes with visual depth, visual coverage is not fully consistent with the question, and a capacity fixed before the language model cannot serve every interaction stage. Q-SCOPE builds a dual-depth regional basis, edits it with query-guided exchange, and contracts the nested set with question-to-visual attention. On the completed LLaVA-1.5-7B schedule that removes 94.4% of the visual token-layer budget, the six-task relative mean is

95.66%. TextVQA drops while POPE and GQA rise. The remaining budgets, transfer models, video benchmarks, and latency measurements are left blank until they are run under the same protocol.

LLM Usage Statement

A large language model assisted English drafting, LaTeX layout, and checks that the wording matched the recorded measurements. The experimental numbers come from the archived Q-SCOPE records. Unmeasured table cells were left blank. The authors checked the method definitions, citation order, and scientific claims.

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